

STAT 525

Chapter 4

Miscellaneous Topics

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individual
CI

$$\Pr(L_i \leq \theta_i \leq U_i) \geq 95\%$$

for every i

A_i

Overall
CI

$$\Pr(L_i \leq \theta_i \leq U_i, \text{ for all } i \dots) \geq 95\%$$

$\bigcap_{i=1}^n A_i$

Simultaneous Inference

Prob. all CIs simultaneously cover the true value $< 95\%$

- Consider a collection of CIs / hypothesis tests
- Each interval has $(1 - \alpha)\%$ confidence level
- What about the overall confidence level?
 - Level of confidence that all constructed intervals contain their true parameter values
 - Often much lower than individual $(1 - \alpha)\%$ level $\rightarrow$ ** When you construct many CIs or run many hypothesis tests, chance that all of them are correct is much lower the chance that one interval/test is correct.*
- Require to adjust individual confidence levels
 - $\downarrow$ Otherwise we will underestimate error.
- Recall confidence band $\rightarrow$ widened individual intervals

Overall CI $\Pr(L_i \leq \theta_i \leq u_i, \text{ for all } i \dots) \geq 95\%$

$\bigcap_{i=1}^n A_i$

$\Downarrow$ opposite

$\Pr(\text{At least one CI fails}) < 1 - 95\%$

Bonferroni Adjustment

- Want simultaneous CIs for θ_1 and θ_2
- Let A_1 denote event that CI excludes θ_1
- Let A_2 denote event that CI excludes θ_2
- We choose individual confidence level as $\Pr(A_i) = \alpha_i$
- What is the probability that both events don't occur?

$$\begin{aligned} \Pr(\overline{A_1} \cap \overline{A_2}) &= \Pr[(A_1 \cup A_2)^c] \\ \Pr(A_1 \cup A_2) &= \Pr(A_1) + \Pr(A_2) - \Pr(A_1 \cap A_2) \\ &\leq \Pr(A_1) + \Pr(A_2) \\ &\downarrow \\ \Pr(\overline{A_1} \cap \overline{A_2}) &\geq 1 - (\Pr(A_1) + \Pr(A_2)) \end{aligned}$$

- If $\alpha_1 + \alpha_2 = .05$, then $\Pr(\overline{A_1} \cap \overline{A_2}) \geq 0.95$

Make it stricter ($\sim 97.5\%$)
so family-wise level adds up to 95%

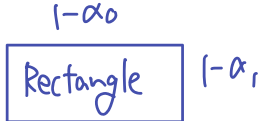
- Want to have family confidence level $1 - \alpha$
- Consider g CIs each using $\alpha_i \equiv \alpha^*$

$$\Pr \left(\bigcap_{i=1}^g \bar{A}_i \right) \geq 1 - g\alpha^*$$

Probability that all succeed

- Use level $1 - \alpha/g$ for each test ($\alpha^* = \alpha/g$)
individual
- Provides lower bound for confidence level
- Increasingly conservative as g increases *less optimal if 'g' too big: interval gets too wide*
- True confidence level often higher than $1 - \alpha$, so larger family-wise can be α used
- A universal method regardless of dependency

Joint Estimation of β_0 and β_1

- Joint confidence Intervals = rectangular confidence region of (β_0, β_1)
- If CIs were independent $\rightarrow$ *In practice, β_0, β_1 are correlated, so this assumption fails*
 - Overall confidence level of rectangle is $(1 - \alpha_0)(1 - \alpha_1)$ 
 - Could set equal to 0.95 and solve for α
- Estimates (b_0, b_1) are not independent and MSE is shared by both CIs.
- Bonferroni Adjustment: $\alpha_0 = \alpha_1 = \alpha/2$
- Given normal error terms, can show (b_0, b_1) multivariate normal, thus can show natural (i.e., smallest) confidence region defined by an ellipse (STAT 524)

Mean Response CIs

- Could apply Bonferroni correction

- Want to know $E(Y|X)$ for g X 's *mean $\bar{Y}$*
- Construct CIs using $\alpha^* = \alpha/g$
- Reasonable approach when g small

$$\hat{Y}_h \pm B \times s(\hat{Y}_h) \text{ where } B = t(1 - \alpha/(2g), n - 2)$$

- Previously discussed Working-Hotelling

- Uses F distribution instead of t distribution
- Coefficient W does not change as g increases *↵ formula for W^2 doesn't contain g .*

$$\hat{Y}_h \pm W \times s(\hat{Y}_h) \text{ where } W^2 = 2F(1 - \alpha, 2, n - 2)$$

Prediction Intervals

- Could apply Bonferroni correction

- Want to know $Y_{h(new)}$ for g X 's new Y
- Construct PIs using $\alpha^* = \alpha/g$
- Reasonable approach when g small

$$\hat{Y}_h \pm B \times s(\text{pred}) \text{ where } B = t(1 - \alpha/(2g), n - 2)$$

- Can also use Scheffé procedure

- Uses F distribution instead of t distribution
- Coefficient S increases as g increases

$$\hat{Y}_h \pm S \times s(\text{pred}) \text{ where } S^2 = gF(1 - \alpha, g, n - 2)$$

Regression through the Origin

- Many instances where the line is known to go through the origin Ex) force v.s. extension

- Statistical model is

no $\beta_0 \rightarrow$ Intercept at (0,0) origin

$$Y_i = \beta_1 X_i + \varepsilon_i \text{ where } \varepsilon_i \sim N(0, \sigma^2)$$

– Can show $b_1 = \sum X_i Y_i / \sum X_i^2$, and $se^2(b_1) = MSE / \sum X_i^2$

– Can fit using option NOINT for MODEL in PROC REG

– !: no more def of β_0

– Degrees of freedoms for some statistics are changed!

$$MSE = \sum (Y_i - \hat{Y}_i)^2 / (n - 1)$$

– Problems with $R^2 = \frac{SSM}{SST}$ $\rightarrow$ based on deviations from $\bar{Y}$, not 0

(0,0) origin:
assuming $X=0, Y=0$.

- But formula for R^2 still uses $\bar{Y}$.
- If true relationship is not $X=0, Y=0$, SSE $\uparrow \uparrow \uparrow$.

- If the line does go through the origin, little is lost fitting a line with both intercept and slope

Rule of thumb: If in doubt, fit both β_0 and β_1 : you lose little.

Measurement Error

- Measurement Error in Y

- Generally not a problem if the measurement error is random, independent with mean zero
- Error term in model represents unexplained variation which is often a combination of many factors not considered *simply adds noise.*

- Measurement Error in X

- Does cause problems
- Often results in biased estimators
- Tends to reduce strength of association
- $Y = \beta_0 + X\beta_1 + \epsilon = \beta_0 + (X^* - \delta)\beta_1 + \epsilon = \beta_0 + X^*\beta_1 + (\epsilon - \delta\beta_1) := \beta_0 + X^*\beta_1 + \epsilon^*$, where X^* and ϵ^* are correlated in general, unless X^* is fixed (Berkson model)

↓
error term becomes
correlated with X :
Breaking regression assumptions

Instead of predicting Y from X ,
sometimes we want to predict X from Y .

Inverse Predictions

- Given Y_h , predict X , such that $Y_h = X_h\beta_1 + \beta_0 + \epsilon$
- Given fitted equation this is

$$\hat{X}_h = \frac{Y_h - b_0}{b_1} \quad b_1 \neq 0$$

- This is the MLE (i.e., function of b_0, b_1)
- **Approximate CI** can be constructed using inverse mapping of CI for new observation Y_h :
Given a known X_h ,

$$b_0 + b_1 X_h - t(1 - \alpha/2, n - 2)s\{pred\} \leq Y_h \leq b_0 + b_1 X_h + t(1 - \alpha/2, n - 2)s\{pred\}$$

$$Y_h - t(1 - \alpha/2, n - 2)s\{pred\} \leq b_0 + b_1 X_h \leq Y_h + t(1 - \alpha/2, n - 2)s\{pred\}$$

$$\implies X_h \in \hat{X}_h \pm t(1 - \alpha/2, n - 2)s\{pred\}/b_1$$

Note: $s\{pred\}$ requires **true X_h** , and we can plug in **$\hat{X}_h$ for an approximation**.

In practice we usually substitute with $\hat{x}_h$ (estimated X_h)

Chapter Review

- Simultaneous Inference / Multiplicity
- Regression through the origin
- Measurement Error
- Inverse predictions